False Position Method

July 2026

1 Introduction

The False Position Method (also known as the Regula Falsi Method) is a numerical root-finding algorithm used to find a solution of the nonlinear equation

$$f(x) = 0.$$

Like the Bisection Method, it requires the function to be continuous on an interval where the function values at the endpoints have opposite signs. However, instead of repeatedly bisecting the interval, the False Position Method approximates the root by using the x-intercept of the secant line joining the two endpoints.

2 Mathematical Foundation

2.1 Linear Interpolation

The linear interpolation formula is used for estimating the value of a function between two known points. Suppose two known points on the function are $(x_0, y_0), (x_1, y_1)$, and the linear interpolant is the straight line between these points. The equation of linear interpolation is given by

$$y = y_0 + \frac{y_1 - y_0}{x_1 - x_0}(x - x_0).$$

where x is the point at which the function value is to be estimated, y is the interpolated value.

2.2 False Position Method

The False Position Method combines linear interpolation with the bracketing strategy used in the Bisection Method. Given an interval $[x_0, x_1]$ such that $f(x_0)f(x_1) < 0$, the equation of the secant line is

$$y = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x_r - x_0).$$

The approximate root is obtained by finding the x-intercept of the secant line. Since the x-intercept satisfies $y = 0$, substituting this condition into the equation gives

$$0 = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x_r - x_0).$$

Solving this equation for x_r gives:

$$x_r = x_0 - \frac{x_1 - x_0}{f(x_1) - f(x_0)}f(x_0).$$

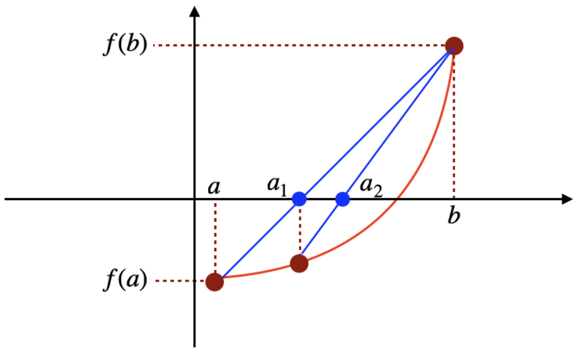

Figure 1: False Position Method

3 Algorithm Flow

At every iteration, the current interval is used to construct a secant line through the two endpoints. The x-intercept of the secant line is taken as the next approximation to the root. Depending on the sign of $f(x_r)$, one endpoint of the interval is replaced while the other remains unchanged. Consequently, the interval continues to bracket the root throughout the iteration.

Algorithm 1 False Position Method

Input: Continuous function $f(x)$, interval $[a, b]$, Tolerance ϵ, Maximum iterations N
Output: Approximate root x_r

```
if f(a)f(b) ≥ 0 then
    return failure
end if
for k = 1 to N do
    x_r = a − (b−a)/(f(b)−f(a)) f(a)
    if |f(x_r) < ε| then
        return x_r
    end if
    if f(a)f(x_r) < 0 then
        b = x_r
    else
        a = x_r
    end if
end for
return failure
```

4 Convergence Analysis

4.1 Convergence

The False Position Method is guaranteed to converge for continuous functions when the initial interval brackets a root. Unlike the Bisection Method, the approximation is obtained from the x-intercept of the secant line rather than the midpoint. As a result, the method often converges faster than the Bisection Method.

4.2 Endpoint Stagnation

For some functions, one endpoint of the interval may remain fixed for many iterations while the other endpoint continues to move toward the root. This phenomenon is known as endpoint stagnation. When it occurs, the convergence becomes significantly slower because the secant line changes very little from one iteration to the next.

4.3 Convergence Rate

The False Position Method exhibits linear convergence. Although it is often faster than the Bisection Method in practice, endpoint stagnation may reduce its efficiency for certain functions.

5 Numerical Example

To illustrate the False Position Method, consider the nonlinear equation

$$f(x) = x^2 - 2.$$

The exact root is

$$x = \sqrt{2} \approx 1.41421356.$$

Choose the initial interval

$$[a, b] = [1, 2].$$

Applying the False Position Method produces the following sequence of approximations.

Table 1: $f(x) = x^2 - 2$

Iteration	x_r	$f(x_r)$
1	1.3333	-0.2222
2	1.4	-0.04
3	1.4117	-0.00692
4	1.4138	-0.00119
$\vdots$	$\vdots$	$\vdots$
11	1.41421	-5.20563×10^{-9}

After 11 iterations, the approximate root is 1.41421.